جامعة خليفة
Khalifa University

Exploring Neutrino Oscillations in KM3NeT

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Course Project for Theoretical Astroparticle Physics

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Agenda

1. History of Neutrinos
2. Neutrino Interactions & Types
3. Pivotal Neutrino Experiments
4. Mass Order Hierarchy
5. The KM3NeT Experiment
6. Broader Physics Research

Neutrino History

Neutrino History

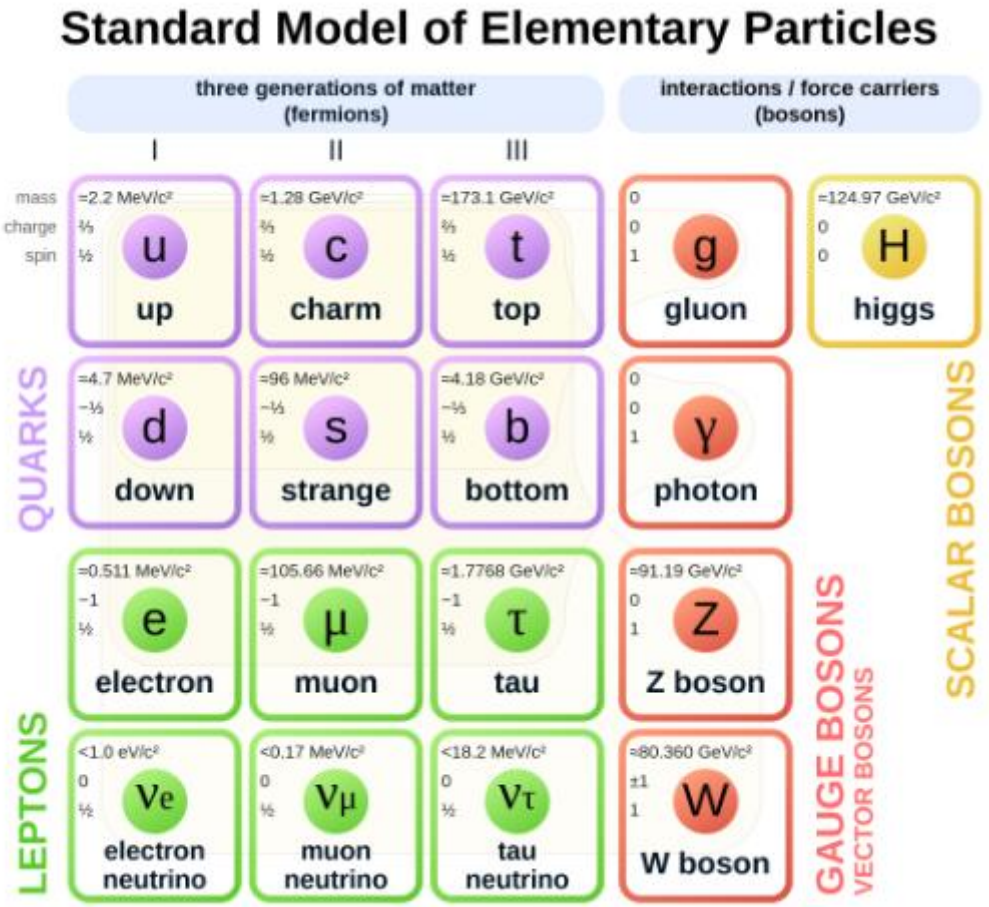
- 1930 — Pauli proposes a neutral, nearly undetectable particle to restore energy–momentum conservation in beta decay.
- 1933 — Fermi develops the theory and names it the “neutrino” (“little neutral one”).
- 1956 — Reines & Cowan detect antineutrinos via inverse beta decay, confirming the particle’s existence.
- Since then — Extensive progress on neutrino properties and interactions.
- Ongoing — Key open questions about neutrinos’ place in the Standard Model and their broader role in physics.



Wolfgang Pauli (1900-1958)

Neutrino's Place in the Standard Model

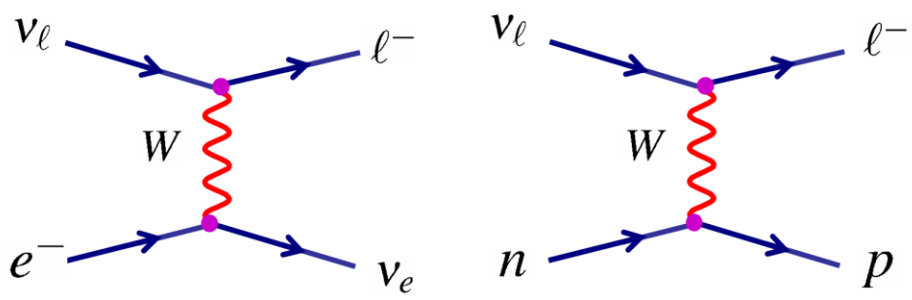
- Leptons with no **electric charge** and no **color charge**
- Do not feel electromagnetic or strong forces; interact via the **weak force** (Z^0, $W^{\pm}$)
- Exist in three flavors: ν_e, ν_μ, ν_τ
- SM predicts no mass, but experiments predict mass.
- Change flavor when propagating.
- Extremely weakly interacting → very small cross sections, highly penetrating
- Can interact via **charged-current** or **neutral-current** processes



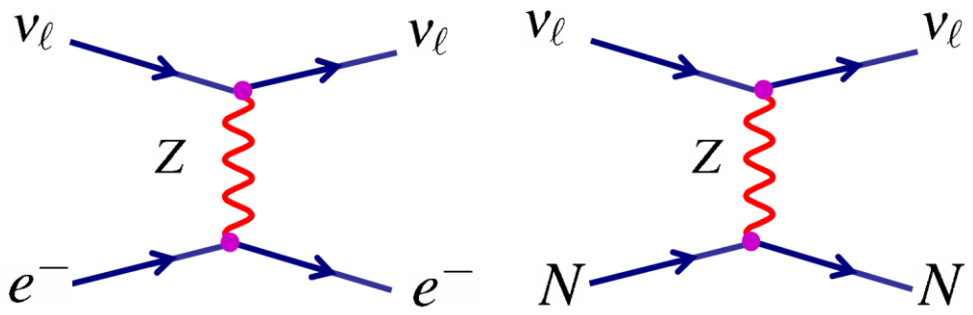
Neutrino Interactions & Types

Neutrino Interactions

Neutrinos interact with matter via the weak interaction in 2 process.



Charged Current



Neutral Current

- Leptons formed from neutrino flavor
- Energy transfers from neutrino to electron/nucleon

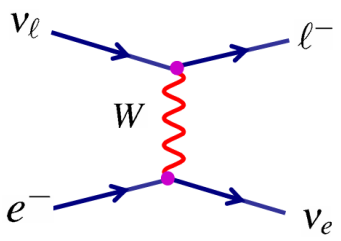
Adapted from [Mark T. 2009]

Neutrino Interactions Threshold

Neutrino detection depend on **neutrino energy** & **flavour**.

Neutrinos may not interact with matter if they have **low energies** (i.e. do not pass E_{th}).

CC Interaction on electrons

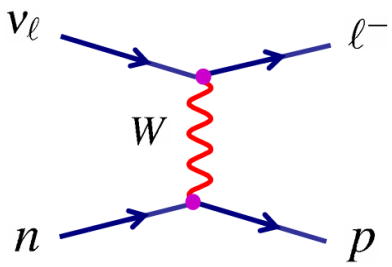


$$s = (p_\nu + p_e)^2 = (E_\nu + m_e)^2 - E_\nu^2$$

Require: $s > m_\ell^2$

$$\Rightarrow \boxed{E_\nu > \left[\left(\frac{m_\ell}{m_e}\right)^2 - 1\right]\frac{m_e}{2}}$$

CC Interaction on nucleons



$$s = (p_\nu + p_n)^2 = (E_\nu + m_n)^2 - E_\nu^2$$

Require: $s > (m_\ell + m_p)^2$

$$\Rightarrow \boxed{E_\nu > \frac{(m_p^2 - m_n^2) + m_\ell^2 + 2m_p m_\ell}{2m_n}}$$

CC interactions are of most importance since they produce **charged leptons** we can detect!

Adapted from [Mark T. 2009]

Neutrino Interactions Threshold

Neutrino detection depend on **neutrino energy** & **flavour**.

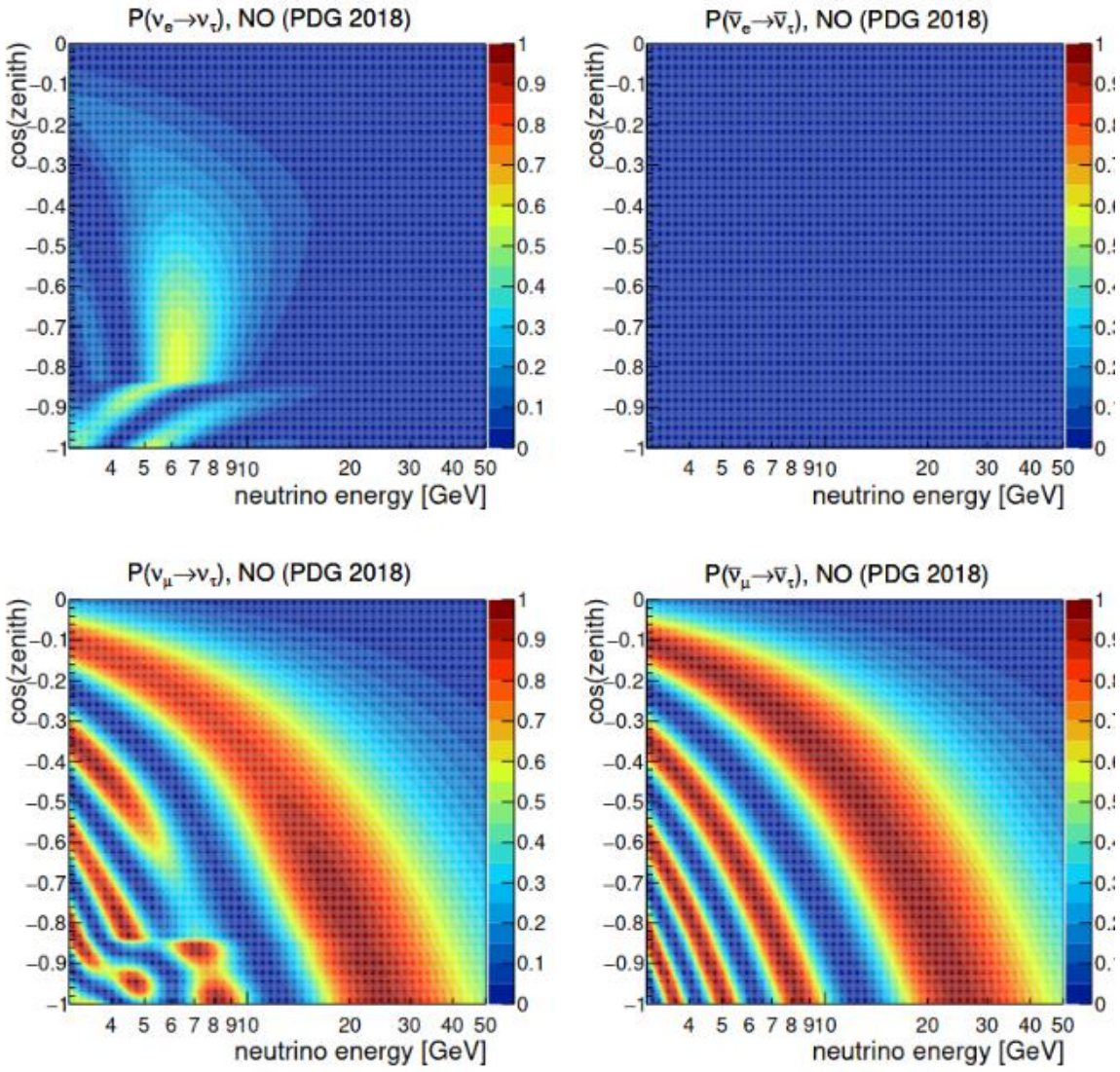
Neutrinos may not interact with matter if they have **low energies** (i.e. do not pass E_{th}).

CC Interaction on electrons	$E_{\nu_e} > 0$	$E_{\nu_\mu} > 11\,\mathrm{GeV}$	$E_{\nu_\tau} > 3090\,\mathrm{GeV}$
CC Interaction on nucleons	$E_{\nu_e} > 0$	$E_{\nu_\mu} > 110\,\mathrm{MeV}$	$E_{\nu_\tau} > 3.5\,\mathrm{GeV}$

- Solar ν_e (~ 1 MeV) "disappear" when oscillating to ν_μ or ν_τ
- Atmospheric ν_μ (~ 1 GeV) "disappear" when oscillating to ν_τ

Adapted from [Mark T. 2009]

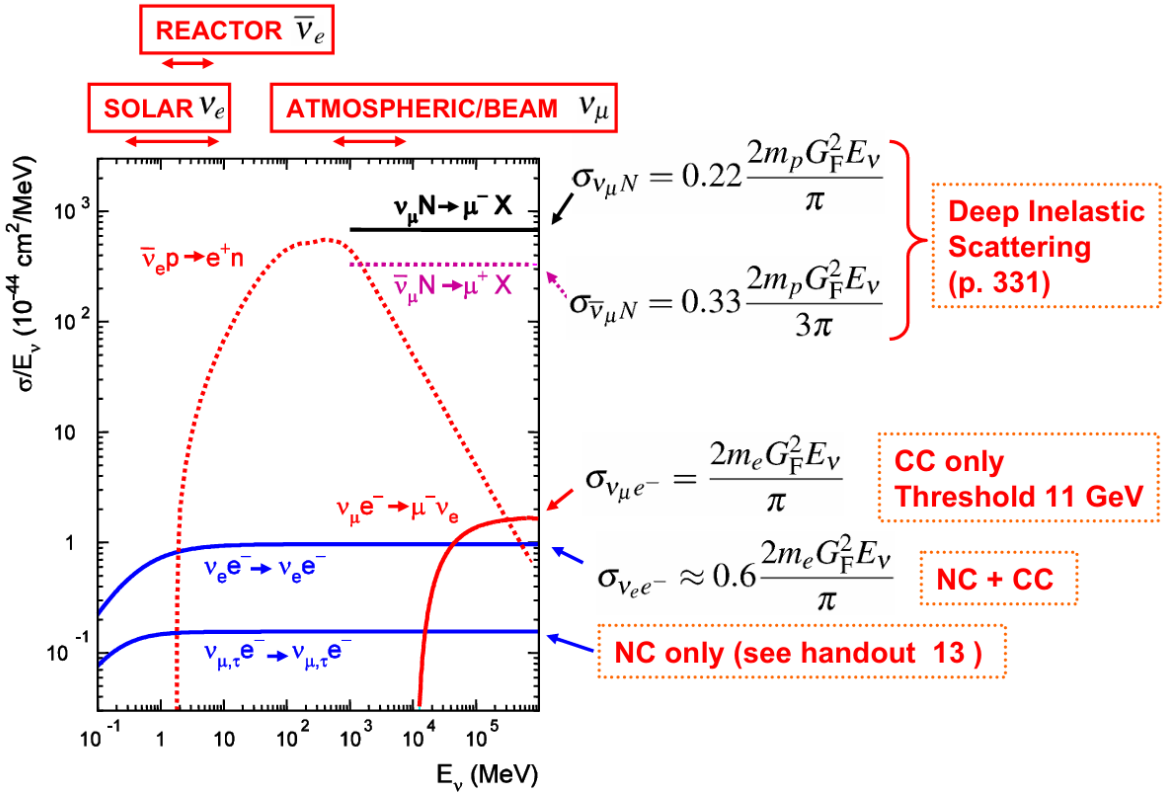
Atmospheric Neutrino Oscillations



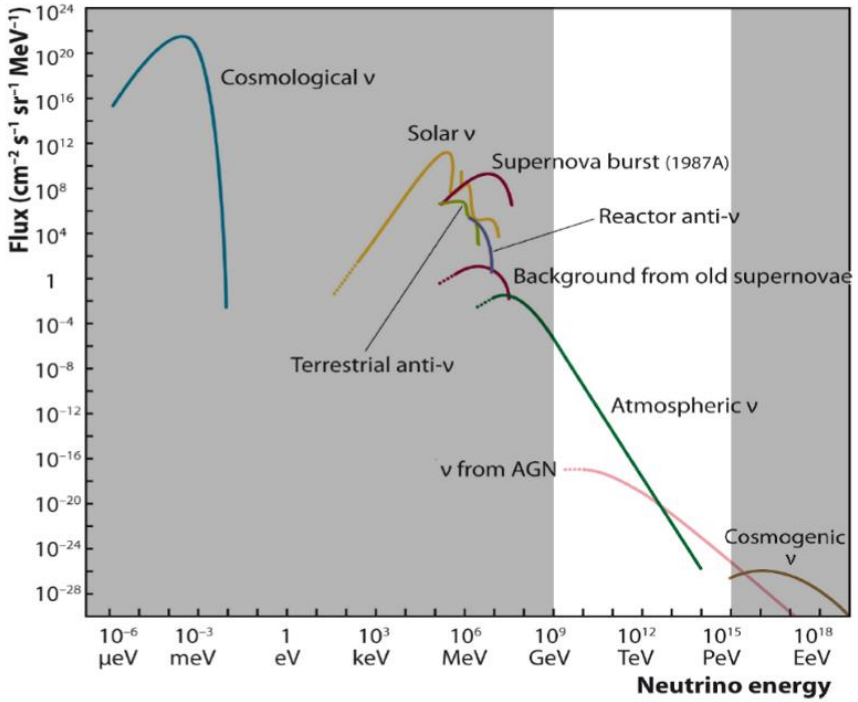
Adapted from [Steffan H. 2021]

Neutrino Detection

Neutrino detection depends on **energy** & **type**.



Crossections of Neutrino Interactions

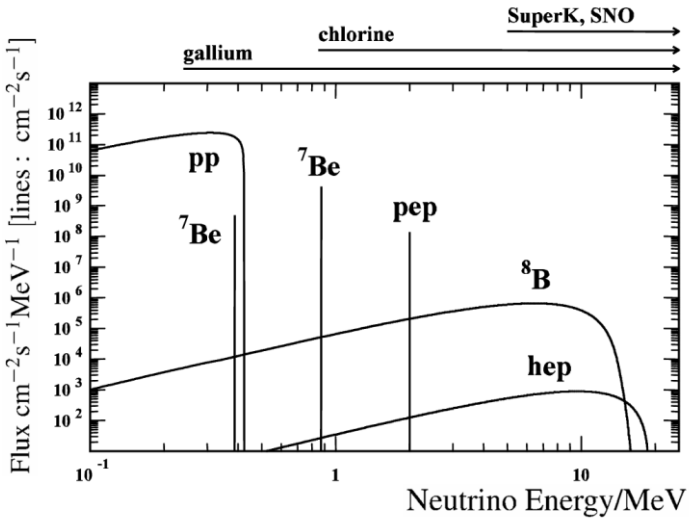


Energy & Flux of Neutrino Types

Solar Neutrinos

The Sun produces a large flux of **electron neutrinos**.

$$2 \times 10^{38}\, \nu_e\, \mathrm{s}^{-1}$$



Many different nuclear reactions lead to a **complex neutrino energy spectrum**.

$p + p \rightarrow d + e^+ + \nu_e$ $\quad E_\nu < 0.5\,\mathrm{MeV}$

$^8B \rightarrow {}^8Be^* + e^+ + \nu_e$ $\quad E_\nu \sim 5\mathrm{MeV}$

$$p + e^- + p \rightarrow d + \nu_e$$
$$^7Be + e^- \rightarrow {}^7Li + \nu_e$$
$$^3He + p \rightarrow {}^4He + e^+ + \nu_e$$

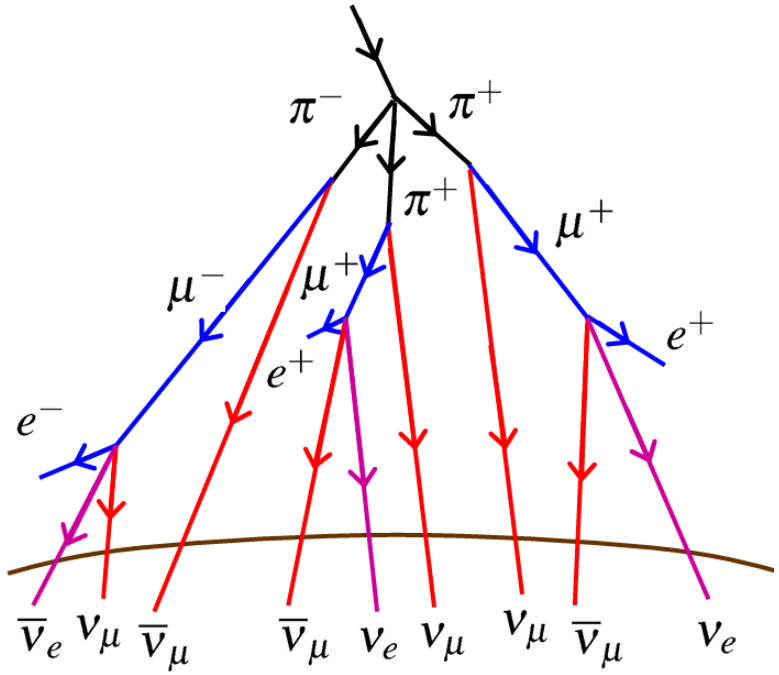
Experiments saw a deficit electron neutrino flux than theorized – **The Solar Neutrino Problem**

Adapted from [Mark T. 2009]

Atmospheric Neutrinos

High Energy Cosmic Rays (up to 10^{20} eV) interact in the upper part of the Earth's atmosphere.

Cosmic Rays are mostly made of (~86% protons, ~ 11% helium, ~ 1% heavier nuclei, ~ 2% electrons)



Neutrinos are mainly produced from

$$\pi^+ \longrightarrow \mu^+ \nu_\mu \qquad \pi^- \longrightarrow \mu^- \overline{\nu}_\mu$$
$$\hookrightarrow e^+ \nu_e \overline{\nu}_\mu \qquad \hookrightarrow e^- \overline{\nu}_e \nu_\mu$$

Neutrino Energy: $E_\nu \sim 1$ GeV

Expected Flavor:

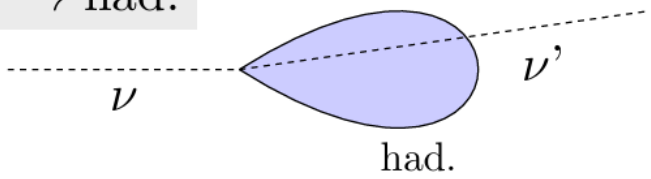
$$\frac{N(\nu_\mu + \overline{\nu}_\mu)}{N(\nu_e + \overline{\nu}_e)} \approx 2$$

Adapted from [Mark T. 2009]

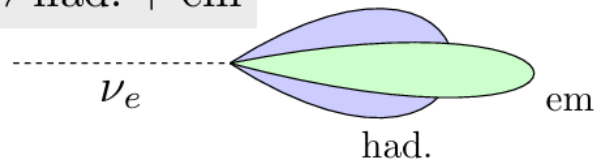
Neutrino Event Topologies

shower-like

$\nu + N \xrightarrow{NC}$ had.

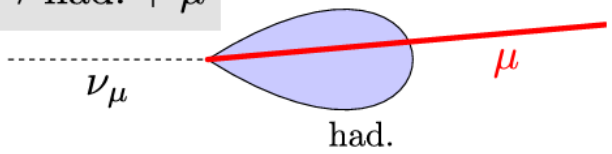


$\nu_e + N \xrightarrow{CC}$ had. + em



track-like

$\nu_\mu + N \xrightarrow{CC}$ had. + μ

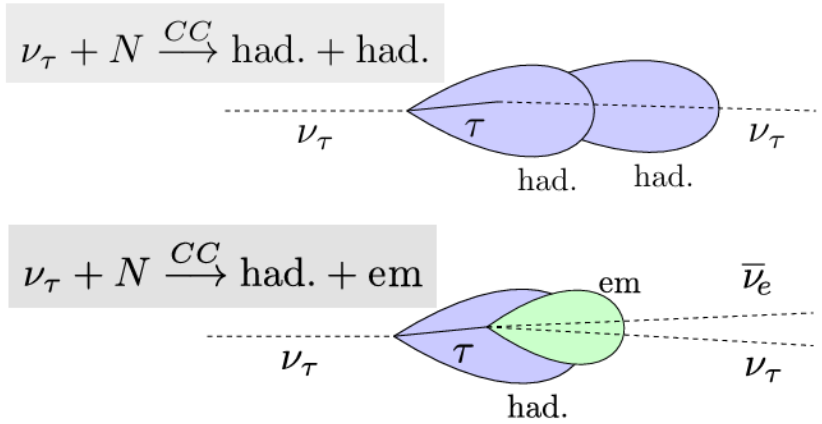


- Neutrinos produce **hadronic showers**.
- **Neutral Current**: Neutrino escapes so light looks point-like.
- **Charged Current**: Neutrino produce leptons and interactions differ depending on neutrino flavor.
  - ν_e produce $e^{\pm}$ (~36 cm) which produce an electromagnetic shower (~ few m) this produce ~20% more light.
  - ν_μ produce μ which has a longer lifespan and has a track-like detection.

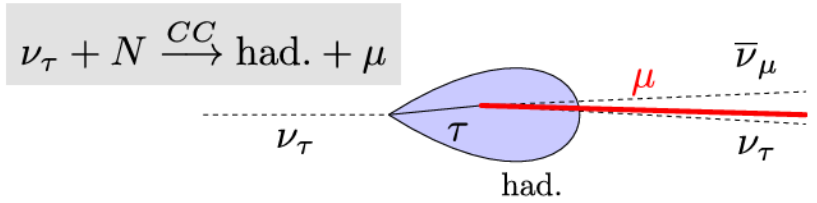
Adapted from [Steffan H. 2021]

Neutrino Event Topologies

shower-like



track-like



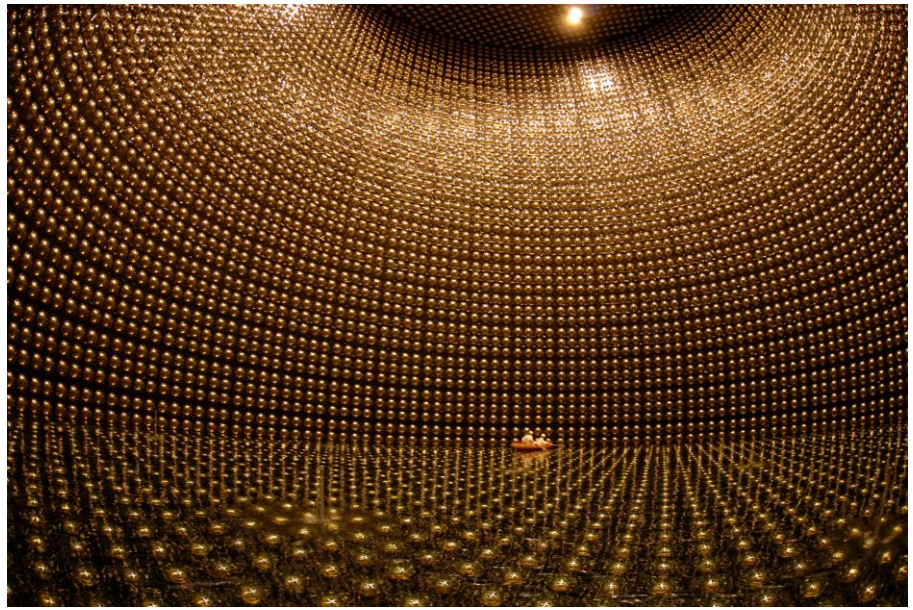
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  - ν_e produce $e^{\pm}$ (~36 cm) which produce an electromagnetic shower (~ few m) this produce ~20% more light.
  - ν_μ produce μ which has a longer lifespan and has a track-like detection.
  - ν_τ produce τ which has such a short lifespan (~ 2.9×10^{-13} s) that it can form either a further hadronic cascade (~ 17%), $e^{\pm}$ producing an em shower (~ 65%), or a μ in a track (~ 17%),.

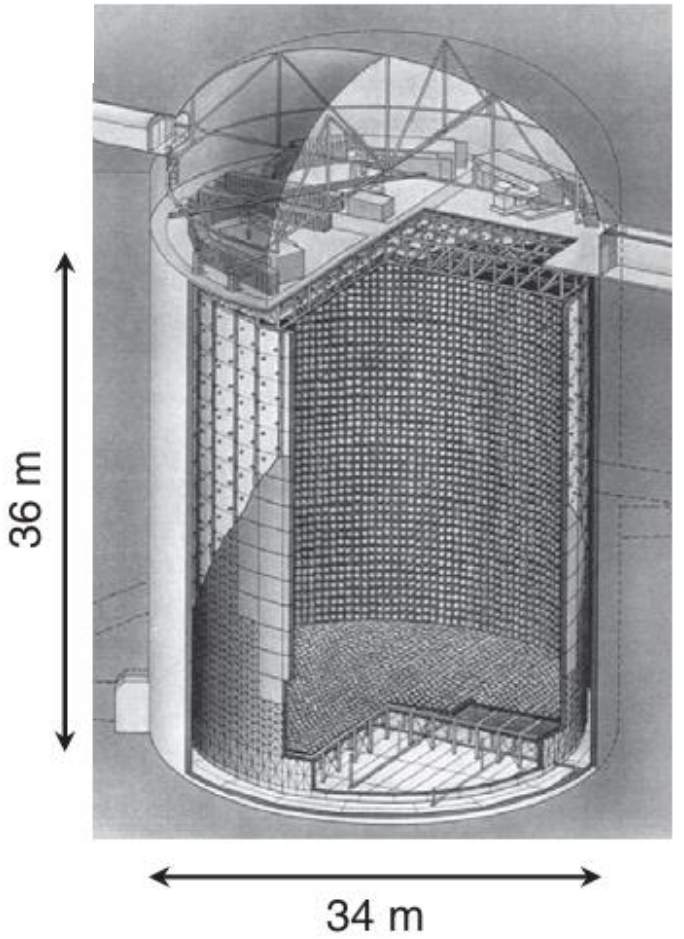
Adapted from [Steffan H. 2021]

Pivotal Neutrino Experiments

Super Kamiokande Experiment

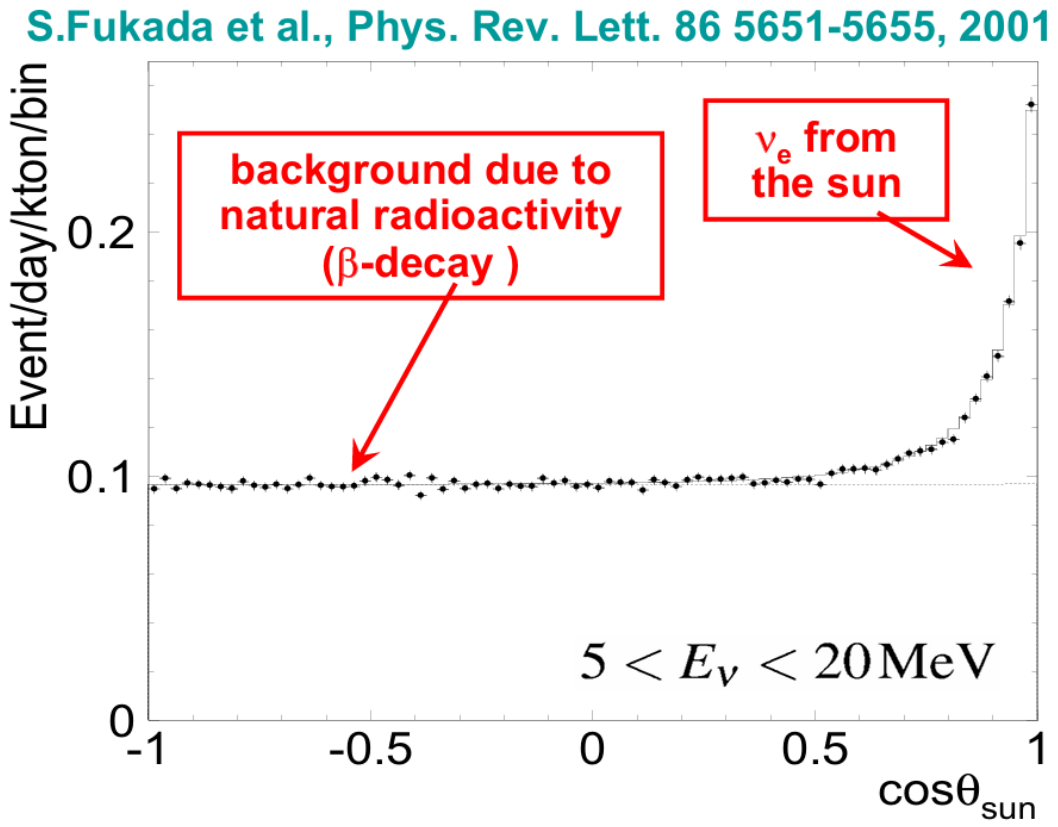
- 50,000-ton water Cherenkov detector (~36 m × 34 m)
- Lined with 11,146 PMTs that can see single photons.
- Deep underground to filter cosmic rays from atmosphere.





Adapted from [Mark T. 2009]

Super Kamiokande Experiment



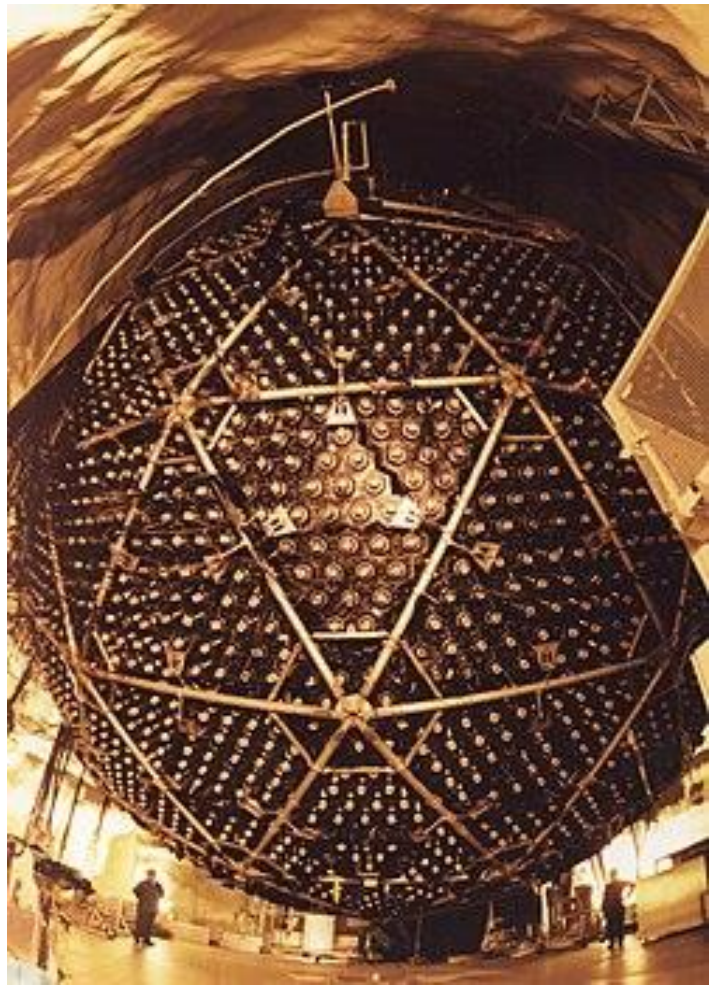
- Evidence of ν_e from the sun.
- However, flux lower than expected!

DATA/SSM = 0.45±0.02

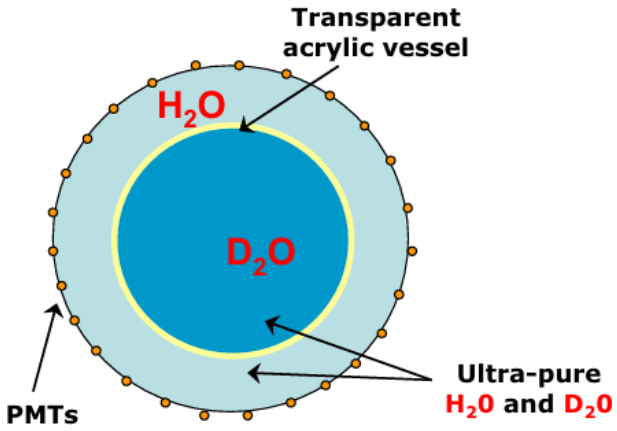
SSM = "Standard Solar Model" Prediction

Adapted from [Mark T. 2009]

Sadbury Neutrino Observatory (SNO)



- SNO is a deep mine in Ontario, Canada.
- Main objective to detect solar neutrinos.
- Recipients of the **2015 Nobel prize** for confirming evidence of neutrino oscillations with super Kamiokande team.

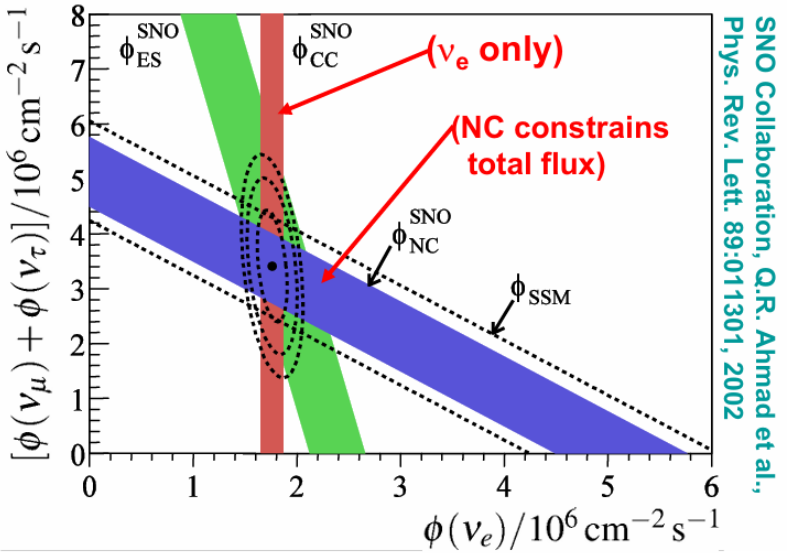


Adapted from [Mark T. 2009]

Sadbury Neutrino Observatory (SNO)

- Different interactions measure different neutrino flavour fluxes.

CC : 1968 ± 61	$\propto \phi(\nu_e)$
ES : 264 ± 26	$\propto \phi(\nu_e) + 0.154[\phi(\nu_\mu) + \phi(\nu_\tau)]$
NC : 576 ± 50	$\propto \phi(\nu_e) + \phi(\nu_\mu) + \phi(\nu_\tau)$



- Evidence of neutrino flux from ν_μ and ν_τ from the sun ⇒ neutrino oscillation!

SNO Results:

$$\phi(\nu_e) = (1.8 \pm 0.1) \times 10^{-6}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

$$\phi(\nu_\mu) + \phi(\nu_\tau) = (3.4 \pm 0.6) \times 10^{-6}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

SSM Prediction:

$$\phi(\nu_e) = 5.1 \times 10^{-6}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

Adapted from [Mark T. 2009]

Neutrino Mass Hierarchy

Neutrino Masses

- Neutrinos were speculated to have no mass by **SM**.
- However, neutrino oscillations require **non-zero mass** neutrinos.
- We can only measure the **mass limits**.

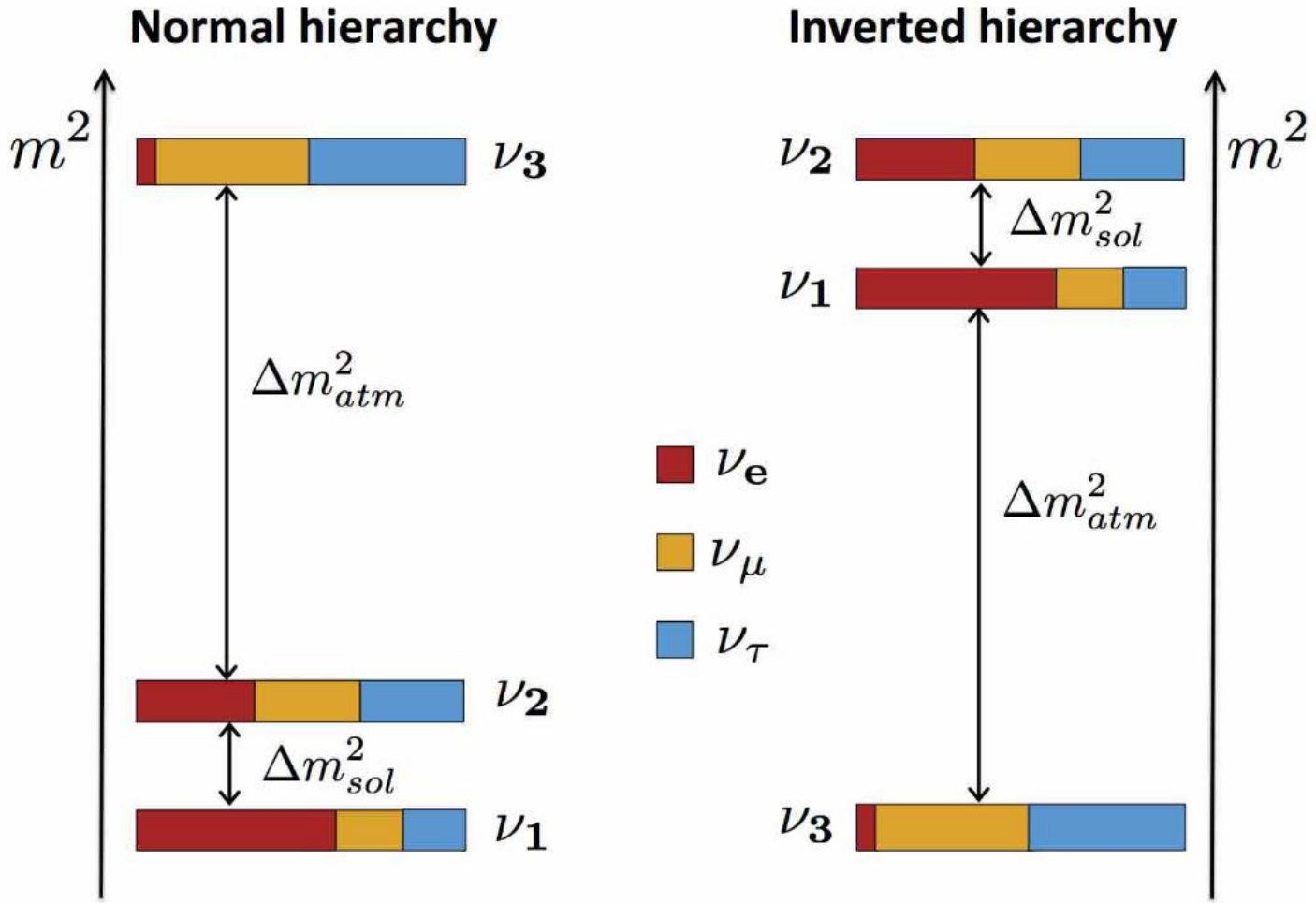
$$m_\nu(e) < 2\,\mathrm{eV}; \quad m_\nu(\mu) < 0.17\,\mathrm{MeV}; \quad m_\nu(\tau) < 18.2\,\mathrm{MeV}$$

- Neutrino experiments can only determine:

$$|\Delta m^2_{ji}| = |m^2_j - m^2_i|$$

  - Atmospheric neutrino oscillations: $|\Delta m^2|_{\mathrm{atmos}} \sim 2.5 \times 10^{-3}\,\mathrm{eV}^2$
  - Solar neutrino oscillations $|\Delta m^2|_{\mathrm{solar}} \sim 8 \times 10^{-5}\,\mathrm{eV}^2$

Mass Order Hierarchy



Neutrino States

- A neutrino state can be written as a linear combination of its flavor or mass states.

$$|\nu\rangle = \sum_{\alpha=e,\mu,\tau} c_\alpha |\nu_\alpha\rangle$$

- There must be a **matrix transformation** that takes you from the flavor basis to the mass basis.

$$\begin{pmatrix} |\nu_e\rangle \\ |\nu_\mu\rangle \\ |\nu_\tau\rangle \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} |\nu_1\rangle \\ |\nu_2\rangle \\ |\nu_3\rangle \end{pmatrix} .$$

- This transformation is called the **PMNS matrix** (Pontecorvo-Maki-Nakagawa-Sakata) and must be unitary to conserve probability.

$$U^\dagger U = I$$

PMNS Matrix

- The PMNS matrix is usually expressed in term of 3 rotation angles θ_{12}, θ_{13} and θ_{23} with a complex phase δ, using notation $c_{ij} = \cos(\theta_{ij})$ and $s_{ij} = \sin(\theta_{ij})$

$$\begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}} \times \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{-i\delta} & 0 & c_{13} \end{pmatrix} \times \underbrace{\begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}}$$

Dominates: **"Atmospheric"** **"Solar"**

- Multiplying out:

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}$$

- There are **six** parameters of interest to be measured by experiments:

$\lvert\Delta m_{21}\rvert^2 = \lvert m_2^2 - m_1^2\rvert$	θ_{12}	**Solar and reactor neutrino experiments**
$\lvert\Delta m_{32}\rvert^2 = \lvert m_3^2 - m_2^2\rvert$	θ_{23}	**Atmospheric and beam neutrino experiments**
	θ_{13}	**Reactor neutrino experiments + future beam**
	δ	**Future beam experiments**

Neutrino Oscillation Probability

- Through solving the unitary conditions, taking the PMNS matrix to be real and making approximations of $|\Delta m^2_{31}| \approx |\Delta m^2_{32}|$ One can show that:

$$P(\nu_e \to \nu_e) \approx 1 - 4U_{e1}^2 U_{e2}^2 \sin^2\Delta_{21} - 4(1-U_{e3}^2)U_{e3}^2 \sin^2\Delta_{32}$$
$$P(\nu_\mu \to \nu_\mu) \approx 1 - 4U_{\mu1}^2 U_{\mu2}^2 \sin^2\Delta_{21} - 4(1-U_{\mu3}^2)U_{\mu3}^2 \sin^2\Delta_{32}$$
$$P(\nu_\tau \to \nu_\tau) \approx 1 - 4U_{\tau1}^2 U_{\tau2}^2 \sin^2\Delta_{21} - 4(1-U_{\tau3}^2)U_{\tau3}^2 \sin^2\Delta_{32}$$
$$P(\nu_e \to \nu_\mu) = P(\nu_\mu \to \nu_e) \approx -4U_{e1}U_{\mu1}U_{e2}U_{\mu2} \sin^2\Delta_{21} + 4U_{e3}^2 U_{\mu3}^2 \sin^2\Delta_{32}$$
$$P(\nu_e \to \nu_\tau) = P(\nu_\tau \to \nu_e) \approx -4U_{e1}U_{\tau1}U_{e2}U_{\tau2} \sin^2\Delta_{21} + 4U_{e3}^2 U_{\tau3}^2 \sin^2\Delta_{32}$$
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$$\Delta_{21} = \frac{(m_2^2 - m_1^2)L}{4E} = \frac{\Delta m_{21}^2 L}{4E}$$

$$\Delta_{21} = (\phi_2 - \phi_1)/2$$

- The wavelengths associated with $\sin^2\Delta_{21}$ and $\sin^2\Delta_{32}$ are:

"SOLAR" $\lambda_{21} = \frac{4\pi E}{\Delta m_{21}^2}$ "Long"-Wavelength

and

$\lambda_{32} = \frac{4\pi E}{\Delta m_{32}^2}$ "ATMOSPHERIC" "Short"-Wavelength

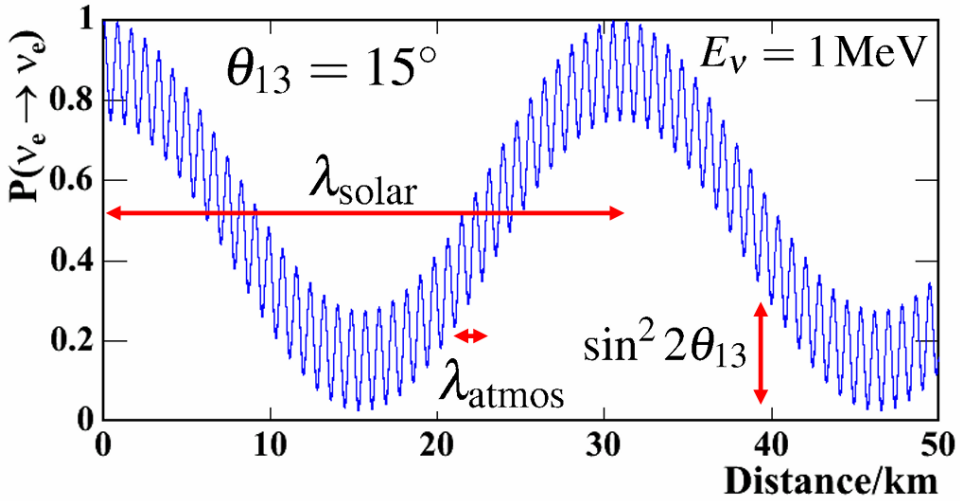
Electron Neutrino Survival Probability

- We can now substitute in the values of the PMNS matrix U_{e1}, U_{e2}, U_{e3} in the **electron neutrino survival probability** get the equation:

$$P(\nu_e \rightarrow \nu_e) \approx 1 - \cos^4\theta_{13}\sin^2 2\theta_{12}\sin^2\Delta_{21} - \sin^2 2\theta_{13}\sin^2\Delta_{32}$$

- The mixing angles and phase differences can be approximated by experiments.
- For 1 MeV neutrino:

$$\lambda_{21} = 30.0\,\text{km}$$
$$\lambda_{32} = 0.8\,\text{km}$$



The KM3NeT Collaboration

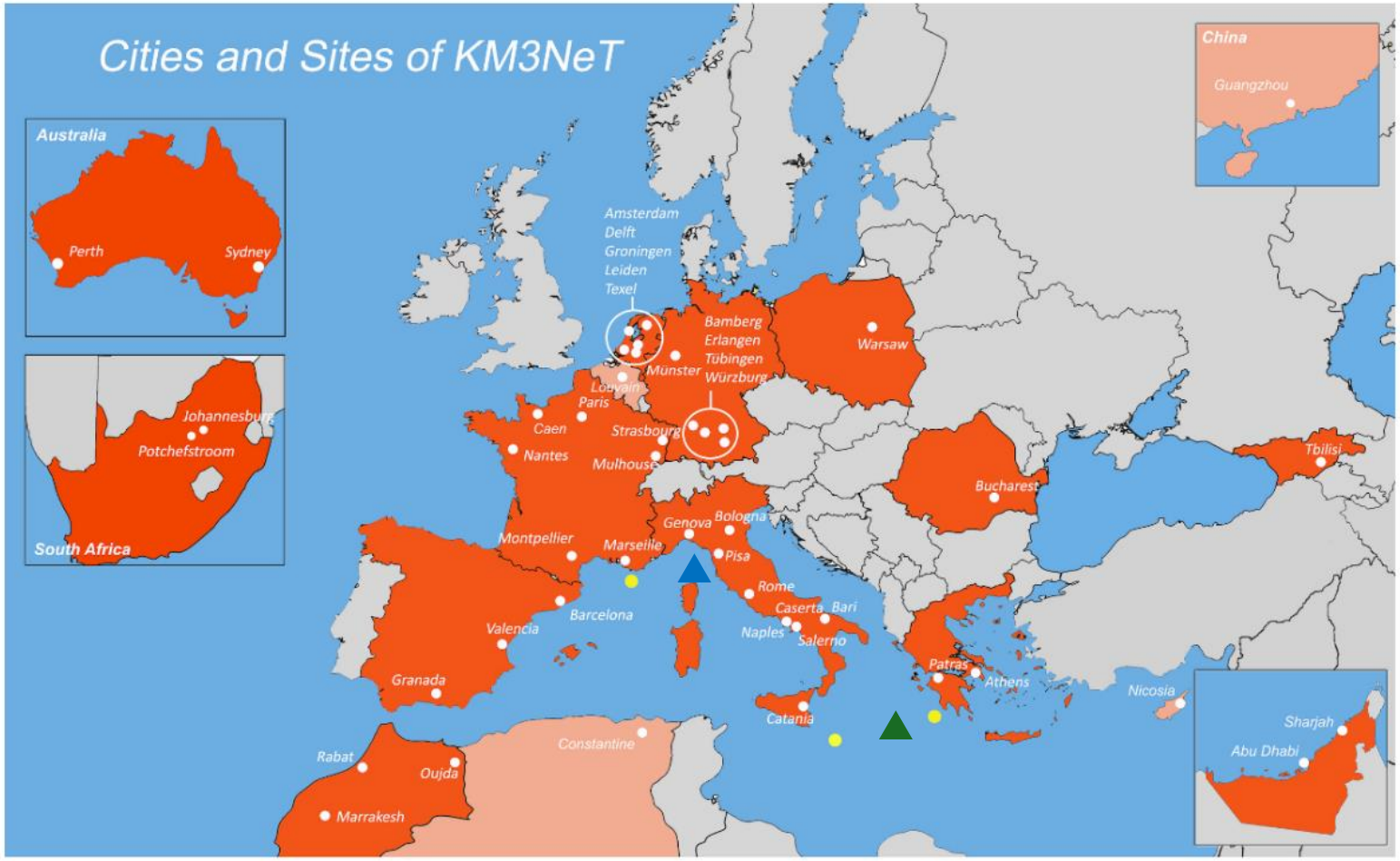
KM3NeT Experiment

International Collaboration with:

- ~ 250 members
- 45 partner institutions
- Over 14 countries

Two detectors at different locations with different focus:

- **ARCA**: Astroparticle Research with Cosmics in the Abyss
- **ORCA**: Oscillation Research with Cosmics in the Abyss



Adapted from [KM3NeT 2016]

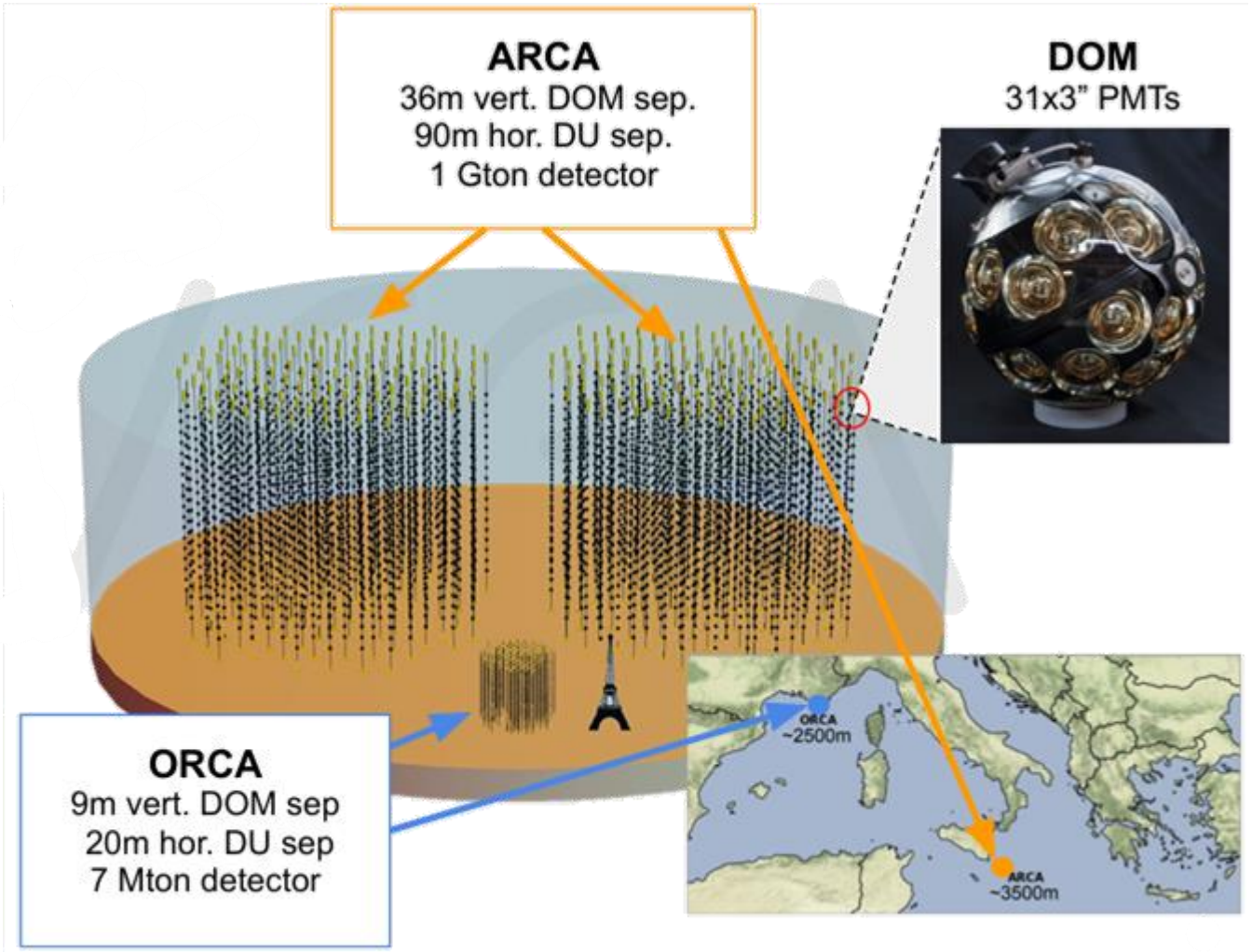
KM3NeT Experiment

KM3NeT/ORCA:

- Low energies (few GeV to 10^2 GeVs)
- Fundamental neutrino property studies
- **Full ARCA:** 115 DUs, 18 DOMs per DU
- **Current ARCA:** 23 DUs deployed

KM3NeT/ARCA:

- High energies (sub-TeV to a few PeV)
- Astrophysical studies
- **Full ARCA:** 230 DUs, 18 DOMs per DU
- **Current ARCA:** 28 DUs deployed



Adapted from [Jorge G. 2024]

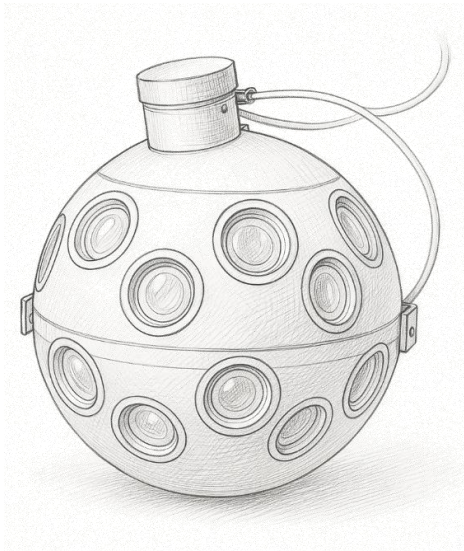
How Are Neutrinos Detected?

1. **Neutrinos Produce Charged Leptons:**
   A neutrino occasionally interacts in/near the seawater, producing a charged lepton (mostly muons from ν_μ). arXiv

2. **Cherenkov Light Produced:**
   That lepton outruns light in water and emits a faint cone of blue Cherenkov photons. arXiv+1

3. **Timing + Geometry track Direction & Energy:**
   By knowing **when** each photon hits (ns precision) and **where** each sensor sits and points, software reconstructs the particle's direction and energy.

💥 **31 × 3″ photomultiplier tubes (PMTs)** turns single photons into electrical pulses. The multi-PMT layout boosts sensitivity and measures "local coincidences" (several PMTs hit together) to reject K^{40} and bioluminescence noise



Digital Optical Module (DOM)

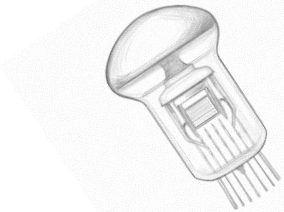
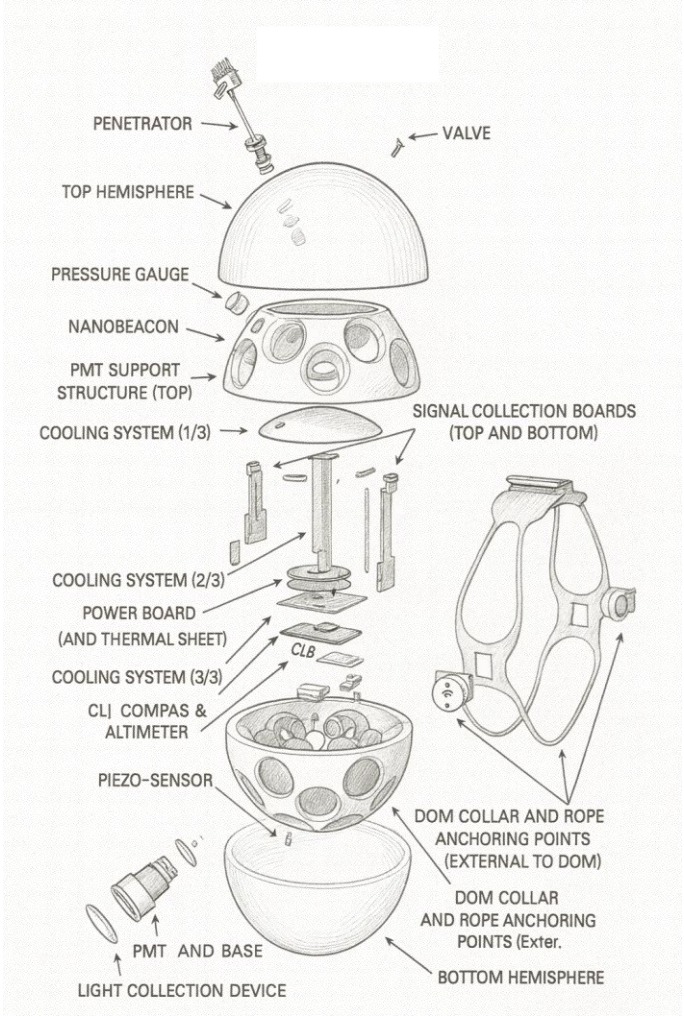


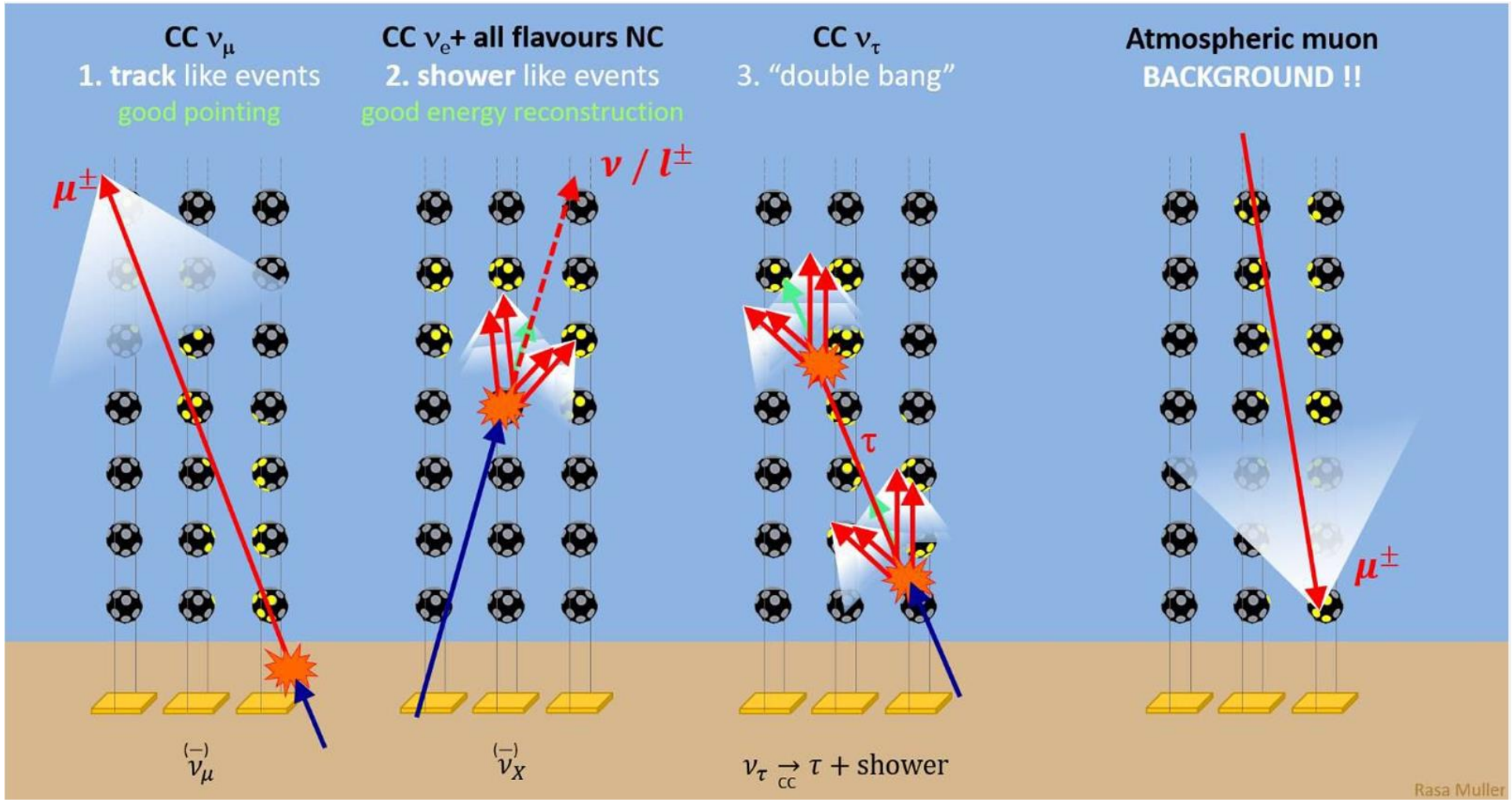
Photo-Multiplier Tubes (PMT)



Complete Model

Adapted from [KM3NeT 2016]

Neutrino Detection Events



Adapted from [Jorge G. 2024]

Oscillation Probabilities

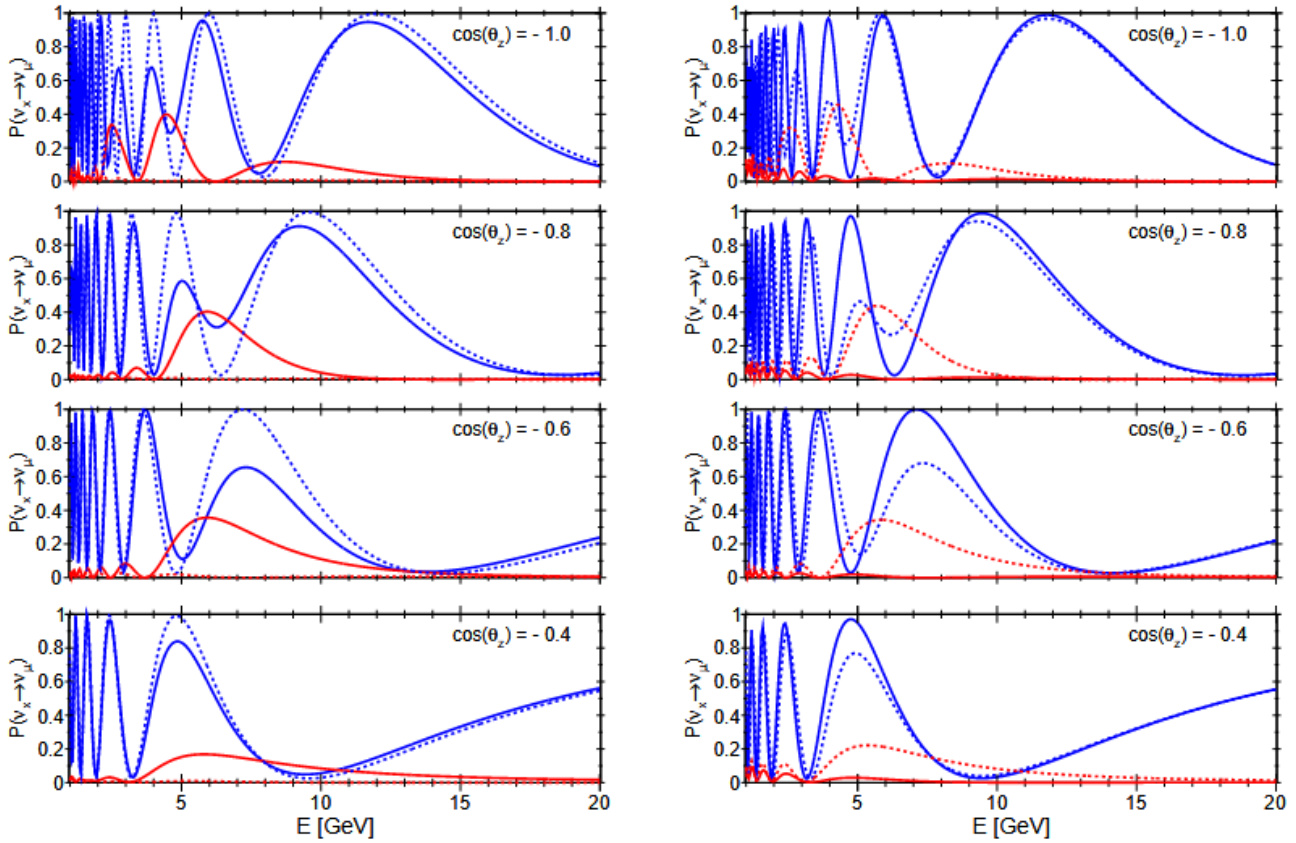


Figure 47: Oscillation probabilities $\nu_\mu \to \nu_\mu$ (blue lines) and $\nu_e \to \nu_\mu$ (red lines) as a function of the neutrino energy for several values of the zenith angle (corresponding to different baselines). The solid (dashed) lines are for NH (IH). For neutrinos (left) and for antineutrinos (right).

from [KM3NeT 2016]

Atmospheric Neutrino Flux

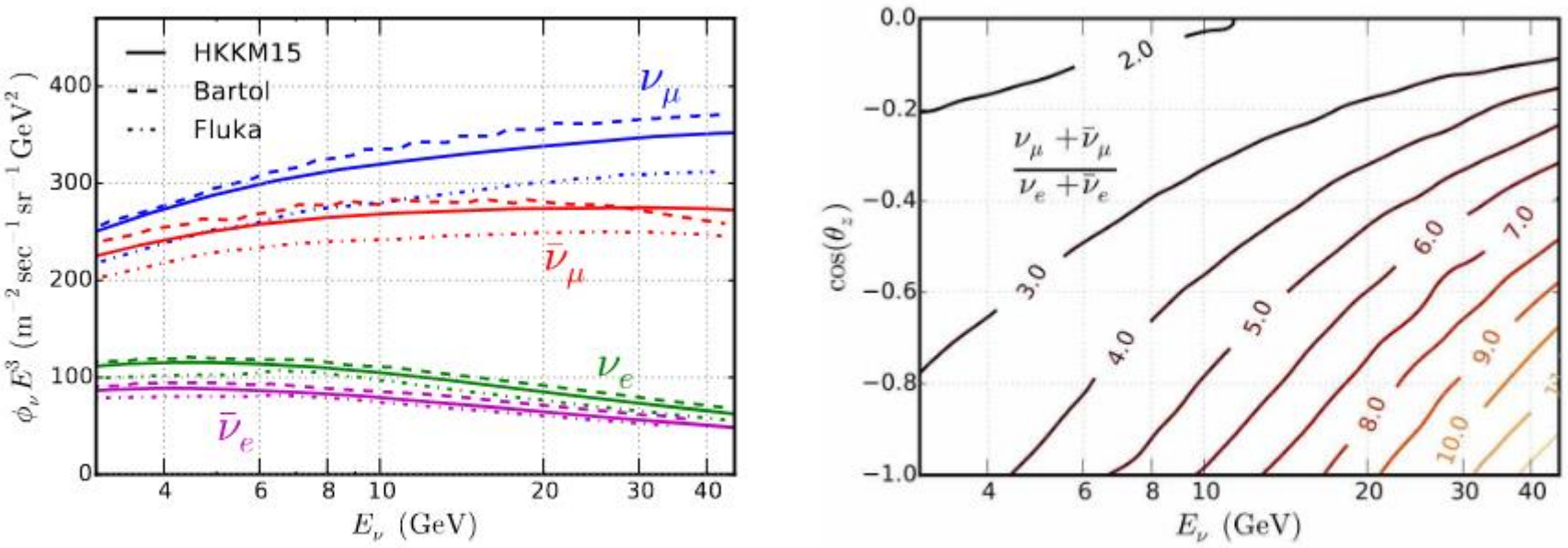


Figure 49: Atmospheric neutrino flux in its different flavour components: absolute values (left) and ratios (right). Taken from [144].

from [KM3NeT 2016]

Neutrino Event Classification

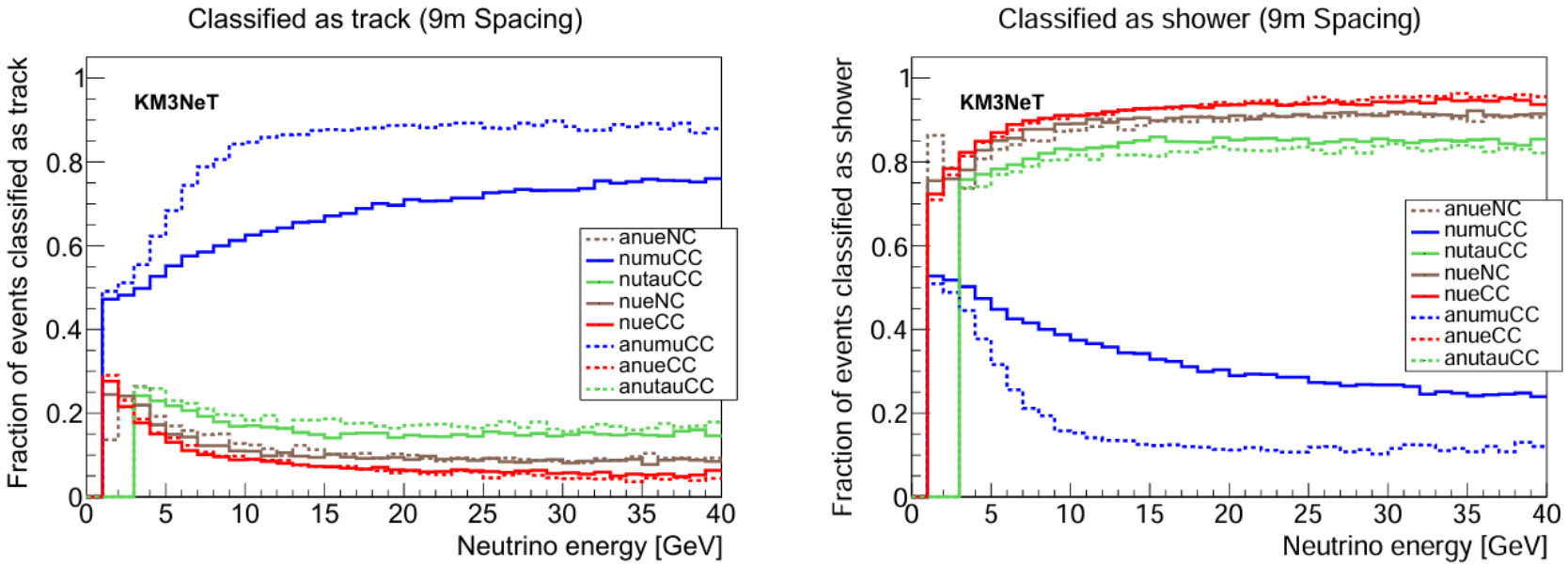


Figure 99: Fraction of events classified as tracks (left) or showers (right) for a detector with 9 m vertical DOM spacing.

from [KM3NeT 2016]

Mixing Angle θ_{23} Significance

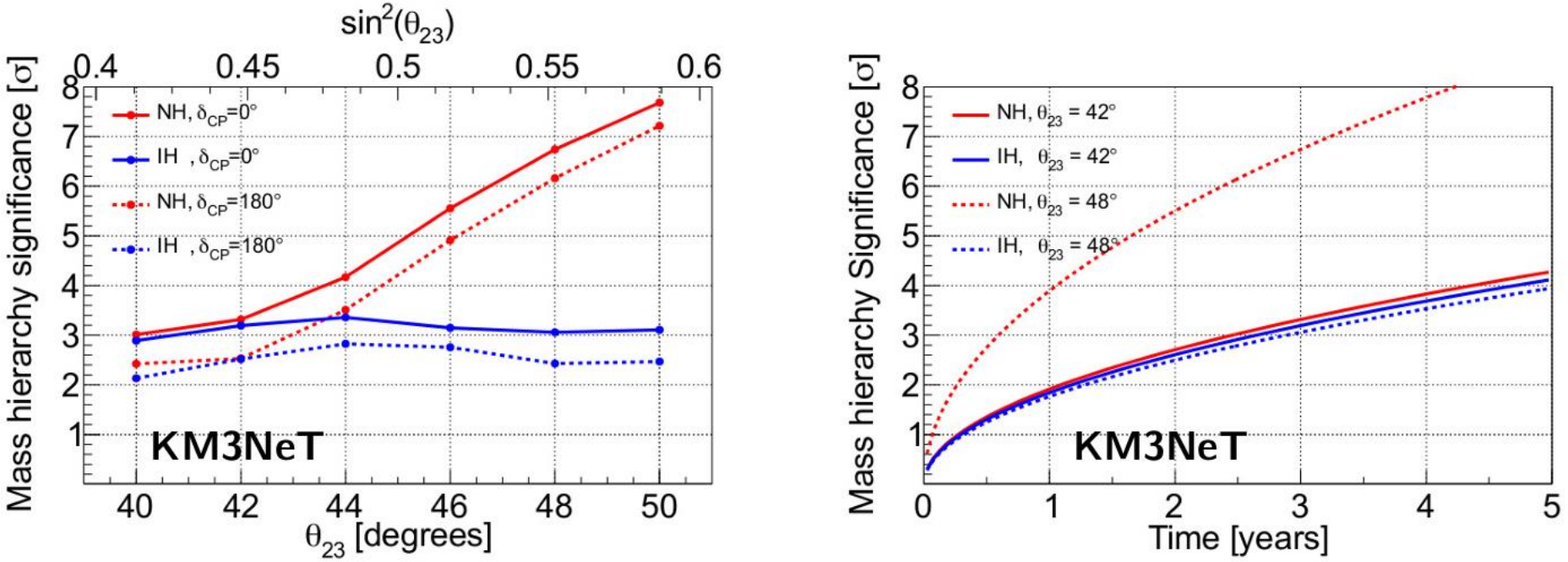
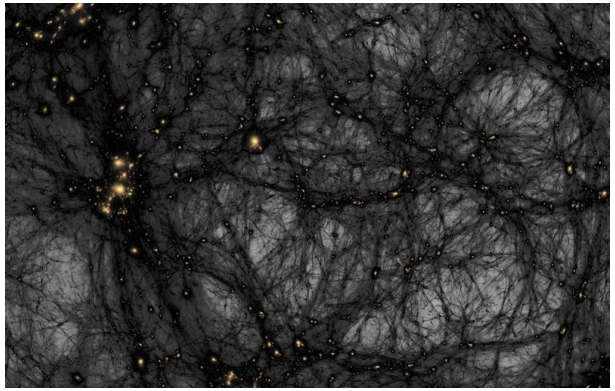


Figure 106: The mass hierarchy sensitivity for 9 m spacing, using the default settings. This includes a fit of θ_{23}, ΔM^2, δ_{CP} and the five systematics. The left plot shows its dependency on θ_{23} for two values δ_{CP} for three years of operation time whereas the right plot illustrates its improvement over time for two selected values of θ_{23} and $\delta_{CP} = 0°$.

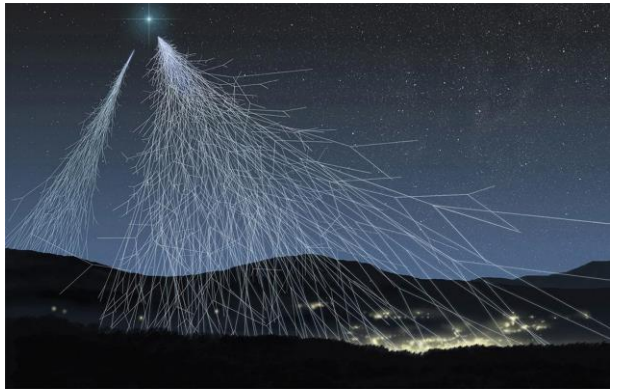
from [KM3NeT 2016]

Broader Implications

Physics Research Opportunities



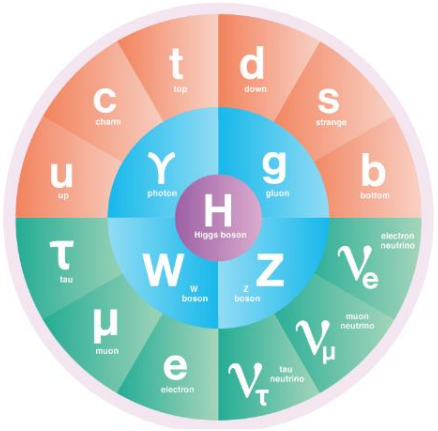
Dark Matter



Cosmic Rays



Gamma Ray Burst (GRBs)



Particle Physics Beyond the Standard Model

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[6] Steffen Hallmann. “Sensitivity to atmospheric tau-neutrino appearance and all-flavour search for neutrinos from the Fermi Bubbles with the deep-sea telescopes KM3NeT/ORCA andANTARES”. en. Available under CC BY-NC-SA 4.0. Doctoral thesis. Erlangen, Germany: Friedrich-Alexander-Universität Erlangen-Nürnberg, Feb. 2021. URL: https://open.fau.de/handle/openfau/15749.

Thank You!